

Joint Timing and Carrier Frequency Offset Estimation with Interference Suppression for Multi-Static ISAC Systems

Zhuoli Liu, Hao Zhu, Menglong Wu

Department of EEIS, University of Science and Technology of China, Hefei, Anhui, China

Email: {lzl_1999, zhuhao, wml12138}@mail.ustc.edu.cn

Abstract—Joint timing offset (TO) and carrier frequency offset (CFO) estimation in multi-static ISAC is particularly challenging in the presence of strong, unknown, and spatially colored interference. The resulting problem is nonconvex and requires high-dimensional optimization. We develop a unified estimation framework that inherently exploits the receive array for interference suppression. For the bi-static single signal-of-interest (SOI) case, we derive a single-SOI estimator (SSE) that alternates between 1-D updates of TO and CFO. Practical FFT-based coarse-to-fine implementations are developed to reduce complexity. We then extend to the multi-static case via an iterative signal isolation and refinement (ISIR) strategy that cyclically estimates per-SOI parameters and locally refines all parameters. This approach avoids computationally prohibitive multi-dimensional searches while maintaining estimation accuracy. Numerical examples demonstrate that the proposed methods achieve accurate parameter estimation, strong interference suppression, and stable performance scaling with respect to the number of SOIs.

Index Terms—Array signal processing, CFO and TO estimation, clock asynchronization, interference suppression, unknown spatially colored interference-and-noise.

I. INTRODUCTION

Integrated sensing and communication (ISAC) has emerged as one of the most promising technologies for the sixth-generation (6G) and beyond wireless networks [1], [2]. While early ISAC research mainly focused on single-static systems with co-located, readily synchronized transceivers, 6G is moving toward denser deployments and network cooperation, which has made multi-static ISAC increasingly attractive [3], [4]. By allowing distributed transceivers to jointly transmit and receive sensing signals, multi-static ISAC can improve coverage, target localization accuracy, and robustness against blockage.

However, a fundamental challenge in multi-static ISAC is clock asynchronization among spatially separated transceivers. Without a common clock reference, each transceiver operates with an independent clock, resulting in both timing offset (TO) and carrier frequency offset (CFO) across the network [5]–[7]. These offsets cause time and phase misalignment of the received signals, deteriorating the estimation accuracy of range, Doppler, and related parameters. Therefore, accurate estimation and compensation for the TO and CFO are critical to ensure reliable ISAC operations.

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Recently, several studies have begun to address asynchronous transceivers in ISAC systems [5], [6], [8], [9]. These works have made substantial progress toward addressing the challenges of asynchronous ISAC, and have laid important groundwork for further advancements. However, these approaches often rely on idealized assumptions such as negligible interference or known noise statistics. In practice, dense multi-static deployments inevitably face strong interference from other transmitters, unintended scatterers, and residual multipath, which can lead to severe performance degradations for synchronization methods designed for weakly interfered channels.

Motivated by these observations, this paper investigates joint TO and CFO estimation under spatially colored interference-and-noise for multi-static ISAC systems affected by clock asynchronization. Concretely, we model the interference-and-noise as spatially colored circularly symmetric complex-valued Gaussian random vectors with zero-mean and unknown covariance matrices [10]–[14], and we exploit the receive array for interference-suppression. Our formulation leverages two types of prior information: the known temporal waveforms of the direct-path SOIs between transmitter-receiver pairs, and the known steering vectors determined by the fixed transmitter-receiver geometry. This differs from prior known-waveform parameter estimation that assumes unknown directions-of-arrival (DOAs) (e.g., [10], [15]–[20]) and from works that assume known steering vectors and waveforms but do not account for clock asynchronization [11]–[13], [21], [22]. In [23], CFO estimation was studied under known steering vectors and temporal waveforms, while TO estimation was not considered. By exploiting the known time-domain waveform information, we estimate both TO and CFO while explicitly modeling the interference statistics, thereby achieving effective interference suppression. This makes the proposed method suitable for dense multi-static environments with interferences.

Notation: We use bold lowercase letters for vectors and bold uppercase letters for matrices. $(\cdot)^H$ denotes the Hermitian transpose. $\odot$ denotes the Hadamard product. $\setminus$ denotes removing an element from a set.

II. PROBLEM FORMULATION

We first consider a single SOI observed by an M -sensor array over L snapshots in a bi-static ISAC system. The

LoS path between a transmitter–receiver pair is exploited to estimate TO and CFO. The received signal is modeled as:

$$\mathbf{x}(l) = \mathbf{a}\gamma s(l - \tau_o)e^{j2\pi f_o l} + \mathbf{e}(l), \quad l = 1, \dots, L, \quad (1)$$

where $\mathbf{x}(l) \in \mathbb{C}^{M \times 1}$ is the l -th data vector, $\mathbf{a} \in \mathbb{C}^{M \times 1}$ denotes the known steering vector of the SOI, $\gamma \in \mathbb{C}$ is its unknown complex amplitude, $s(\cdot)$ is a known probing waveform, $f_o \in [-1/2, 1/2]$ is the unknown normalized CFO, and $\tau_o \in [0, L]$ denotes the unknown delay of the direct-path SOI relative to the receiver, which jointly accounts for the propagation delay and TO. Since the LoS propagation delay is known from geometry, the actual TO can be readily obtained.

The interference-and-noise vectors $\{\mathbf{e}(l)\}_{l=1}^L$ are modeled as temporally independent, spatially colored, identically distributed circularly symmetric complex-valued Gaussian vectors with a zero-mean and unknown covariance matrix $\mathbf{Q}$, i.e.,

$$\mathbb{E}[\mathbf{e}(l_1)\mathbf{e}^H(l_2)] = \mathbf{Q}\delta_{l_1, l_2}, \quad (2)$$

where $l_1, l_2 \in \{1, 2, \dots, L\}$, $\delta_{i,j}$ is the Kronecker delta function. The average power of the SOI waveform is

$$P_s(\tau_o) = \frac{1}{L} \sum_{l=1}^L |s(l - \tau_o)|^2. \quad (3)$$

We define the signal-to-noise ratio (SNR) by treating the interference-and-noise as a generalized term:

$$\text{SNR} = \frac{M|\gamma|^2 P_s(\tau_o)}{\text{tr}(\mathbf{Q})}. \quad (4)$$

Our objective is to estimate γ , f_o , τ_o , and $\mathbf{Q}$ from $\{\mathbf{x}(l)\}_{l=1}^L$.

III. ITERATIVE ESTIMATOR FOR THE BI-STATIC CASE

A. Derivation of the Iterative Estimator

Since $\{\mathbf{e}(l)\}_{l=1}^L$ are i.i.d. circularly symmetric complex-valued spatially colored Gaussian random vectors with covariance matrix $\mathbf{Q}$, i.e., $\mathbf{e}(l) \sim \mathcal{CN}(0, \mathbf{Q})$, the log-likelihood function of the received signal vectors $\{\mathbf{x}(l)\}_{l=1}^L$ can be expressed as (ignoring the constant terms):

$$-L \ln |\mathbf{Q}| - \sum_{l=1}^L \|\mathbf{x}(l) - \mathbf{a}\gamma s(l - \tau_o)e^{j2\pi f_o l}\|_{\mathbf{Q}^{-1}}^2, \quad (5)$$

where $\|\mathbf{y}\|_{\mathbf{Q}^{-1}}^2 \triangleq \mathbf{y}^H \mathbf{Q}^{-1} \mathbf{y}$. We obtain the maximum-likelihood (ML) optimization metric for the problem based on the log-likelihood function. The estimates $\hat{\gamma}$, $\hat{f}_o$, $\hat{\tau}_o$ can be obtained by minimizing the following cost function:

$$\arg \min_{\gamma, f_o, \tau_o} \left| \sum_{l=1}^L [\mathbf{x}(l) - \mathbf{a}\gamma s(l - \tau_o)e^{j2\pi f_o l}] \cdot [\mathbf{x}(l) - \mathbf{a}\gamma s(l - \tau_o)e^{j2\pi f_o l}]^H \right|. \quad (6)$$

This optimization problem is nonconvex and requires high-dimensional search. We address this problem via an alternating optimization approach.

The estimator iterates between estimating the two parameters, i.e., CFO and TO. Each iteration involves two steps. First, f_o is estimated using $\hat{\tau}_o$ in the previous iteration:

$$\hat{f}_o^{(i)} = \arg \min_{f_o} \frac{-|\mathbf{a}^H \hat{\mathbf{R}}^{-1} \mathbf{h}(f_o, \hat{\tau}_o^{(i-1)})|^2}{P_s(\hat{\tau}_o^{(i-1)}) - \mathbf{h}^H(f_o, \hat{\tau}_o^{(i-1)}) \hat{\mathbf{R}}^{-1} \mathbf{h}(f_o, \hat{\tau}_o^{(i-1)})}, \quad (7)$$

where we have defined the sample covariance matrix:

$$\hat{\mathbf{R}} = \frac{1}{L} \sum_{l=1}^L \mathbf{x}(l) \mathbf{x}^H(l), \quad (8)$$

and

$$\mathbf{h}(f_o, \tau_o) = \frac{1}{L} \sum_{l=1}^L \mathbf{x}(l) s^*(l - \tau_o) e^{-j2\pi f_o l}. \quad (9)$$

Then, τ_o is estimated using the $\hat{f}_o$ obtained in the current iteration.

$$\hat{\tau}_o^{(i)} = \arg \min_{\tau_o} \frac{-|\mathbf{a}^H \hat{\mathbf{R}}^{-1} \mathbf{h}(\hat{f}_o^{(i)}, \tau_o)|^2}{P_s(\tau_o) - \mathbf{h}^H(\hat{f}_o^{(i)}, \tau_o) \hat{\mathbf{R}}^{-1} \mathbf{h}(\hat{f}_o^{(i)}, \tau_o)}. \quad (10)$$

After the iteration converges, γ is then estimated using

$$\hat{\gamma} = \frac{\mathbf{a}^H \hat{\mathbf{Q}}^{-1} \mathbf{h}(\hat{f}_o, \hat{\tau}_o)}{P_s(\hat{\tau}_o) \mathbf{a}^H \hat{\mathbf{Q}}^{-1} \mathbf{a}}, \quad (11)$$

where we have defined:

$$\hat{\mathbf{Q}} = \hat{\mathbf{R}} - \frac{\mathbf{h}(\hat{f}_o, \hat{\tau}_o) \mathbf{h}^H(\hat{f}_o, \hat{\tau}_o)}{P_s(\hat{\tau}_o)}. \quad (12)$$

B. Computationally Efficient Implementation

A direct 1-D search over f_o or τ_o is computationally prohibitive. We adopt a coarse-to-fine strategy where the coarse step is implemented via FFTs and the fine step uses a local search (e.g., quadratic-interpolation method).

1) *Coarse estimation of f_o via FFT:* $\mathbf{a}^H \hat{\mathbf{R}}^{-1} \mathbf{h}(f_o, \hat{\tau}_o)$ can be expressed as

$$\mathbf{a}^H \hat{\mathbf{R}}^{-1} \mathbf{h}(f_o, \hat{\tau}_o) = \frac{1}{L} \sum_{l=1}^L \mathbf{a}^H \hat{\mathbf{R}}^{-1} \mathbf{x}(l) s^*(l - \hat{\tau}_o) e^{-j2\pi f_o l}, \quad (13)$$

which is the DFT of $\{\mathbf{a}^H \hat{\mathbf{R}}^{-1} \mathbf{x}(l) s^*(l - \hat{\tau}_o)\}_{l=1}^L$, and it can be implemented by an FFT operation. Similarly, $\mathbf{h}^H(f_o, \hat{\tau}_o) \hat{\mathbf{R}}^{-1} \mathbf{h}(f_o, \hat{\tau}_o)$ can be expressed as

$$\mathbf{h}^H(f_o, \hat{\tau}_o) \hat{\mathbf{R}}^{-1} \mathbf{h}(f_o, \hat{\tau}_o) = \left\| \frac{1}{L} \sum_{l=1}^L \hat{\mathbf{R}}^{-\frac{1}{2}} \mathbf{x}(l) s^*(l - \hat{\tau}_o) e^{-j2\pi f_o l} \right\|_2^2, \quad (14)$$

which is the squared Euclidean norm of the DFT of the $M \times 1$ vectors $\{\hat{\mathbf{R}}^{-\frac{1}{2}} \mathbf{x}(l) s^*(l - \hat{\tau}_o)\}_{l=1}^L$. This transformation can be implemented using M FFT operations.

By evaluating the metric over the DFT grid and picking the minimizing bin, we obtain the coarse estimate of f_o by $M+1$ FFT operations with proper zero paddings to length N_F in parallel.

2) *Coarse estimation of τ_o via FFT*: The function $\mathbf{h}(\hat{f}_o, \tau_o)$ can be expressed as

$$\mathbf{h}(\hat{f}_o, \tau_o) = \frac{1}{L} \sum_{l=1}^L \tilde{\mathbf{x}}(l) s^*(l - \tau_o). \quad (15)$$

where $\tilde{\mathbf{x}}(l) \triangleq \mathbf{x}(l)e^{-j2\pi\hat{f}_ol}$. The resulting $\mathbf{h}(\hat{f}_o, \tau_o) \in \mathbb{C}^{M \times 1}$ can be viewed as the correlation between the scalar sequence $s(l)$ and each component of the vector sequence $\tilde{\mathbf{x}}(l)$. By the correlation theorem, this operation can be implemented efficiently in the frequency domain. By evaluating over all L DFT bins, a coarse TO estimate $\hat{\tau}_o$ is therefore obtained with $2M + 3$ FFTs.

3) *Complexity*: Forming and inverting $\hat{\mathbf{R}}$ entail complexities of $O(M^2L + M^3)$. For each SSE iteration, the dominant computational costs arise from FFT-based coarse estimation: $O((M+1)N_F \log N_F)$ for CFO and $O((2M+3)L \log L)$ for TO.

We refer to this algorithm as single SOI estimator (SSE). It is summarized in Algorithm 1. The fine searches are restricted to the adjacent bins around the coarse estimates. The convergence criterion is the relative change of $\hat{f}_o$ or $\hat{\tau}_o$ is below 10^{-6} .

Algorithm 1 SSE

Input: received signals $\{\mathbf{x}(l)\}_{l=1}^L$, known waveforms $s(t)$, array steering vector $\mathbf{a}$

Output: complex amplitude $\hat{\gamma}$, CFO $\hat{f}_o$ and TO $\hat{\tau}_o$

- 1: Initialize $\hat{\tau}_o^{(i-1)} = 0$, $i = 1$
 - 2: **repeat**
 - 3: Obtain the coarse estimate $\hat{f}_o^{(i)}$ based on $\hat{\tau}_o^{(i-1)}$;
 - 4: Refine $\hat{f}_o^{(i)}$ with 1-D fine search based on the optimization metric in (7);
 - 5: Obtain the coarse estimate $\hat{\tau}_o^{(i)}$ based on $\hat{f}_o^{(i)}$;
 - 6: Refine $\hat{\tau}_o^{(i)}$ with 1-D fine search based on the optimization metric in (10);
 - 7: $i \leftarrow i + 1$;
 - 8: **until** practical convergence
 - 9: $\hat{\gamma} = \frac{\mathbf{a}^H \hat{\mathbf{Q}}^{-1} \mathbf{h}(\hat{f}_o, \hat{\tau}_o)}{P_s(\hat{\tau}_o) \mathbf{a}^H \hat{\mathbf{Q}}^{-1} \mathbf{a}}$.
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IV. EXTENSION TO THE MULTI-STATIC CASE

We now extend our analysis to the case of K SOIs impinging on the array in a multi-static ISAC system. Consider the problem of estimating the complex amplitudes, TOs and CFOs of multiple SOIs arriving at an array of M sensors in the presence of unknown spatially colored interference-and-noise. We assume that the waveforms and steering vectors are known. The received data can be written as:

$$\mathbf{x}(l) = \sum_{k=1}^K \mathbf{a}_k \gamma_k s_k(l - \tau_{o,k}) e^{j2\pi f_{o,k} l} + \mathbf{e}(l), \quad l = 1, \dots, L. \quad (16)$$

The parameter space for this problem has $M^2 + 4 \times K$ unknowns in total. We adopt a cyclic optimization strategy [24], [25] and propose an iterative signal isolation and

refinement (ISIR) approach to efficiently estimate the joint parameters of multiple SOIs.

Let $\mathcal{N}_k = \{(\hat{\gamma}_i, \hat{f}_{o,i}, \hat{\tau}_{o,i}), i = 1, \dots, k\}$ denote the estimated parameters for the first k SOIs. At each iteration k , the algorithm performs two operations: signal isolation for the k -th SOI and refinement of all parameters in the set $\mathcal{N}_k$.

1) *Signal Isolation*: The signal isolation step involves isolating the residual signal corresponding to the k -th SOI by subtracting the contributions from the SOIs we have estimated. When estimating the parameters of the k -th SOI, we treat the contributions of the $(k-1)$ SOIs as known and subtract them from the received data:

$$\mathbf{x}_{\mathcal{N}_{k-1}}(l) = \mathbf{x}(l) - \sum_{i=1}^{k-1} \mathbf{a}_i \hat{\gamma}_i s_i(l - \hat{\tau}_{o,i}) e^{j2\pi \hat{f}_{o,i} l}. \quad (17)$$

This residual signal is then passed to the SSE algorithm for parameter estimation, as in the single SOI case.

2) *Local Refinement*: After obtaining the parameters for the k -th SOI, we refine all parameters in $\mathcal{N}_k$ locally. Concretely, for each parameter tuple $(\gamma_i, f_{o,i}, \tau_{o,i})$ in $\mathcal{N}_k$, we use $\mathbf{x}_{\mathcal{N}_k \setminus (\gamma_i, f_{o,i}, \tau_{o,i})}(l)$ as the measurement signal, and locally refine the parameters similar to the fine search step in SSE. The local refinement step usually converges rapidly in practice.

We summarize our approach in Algorithm 2. In the inner

Algorithm 2 Iterative Signal Isolation and Refinement (ISIR) for Multi-SOI Estimation

Input: received signals $\{\mathbf{x}(l)\}_{l=1}^L$, known waveforms $\mathbf{s}(t)$, number of sources K , array steering matrix $\mathbf{A}$

Output: Estimated parameters: complex amplitude vector $\boldsymbol{\gamma}$, CFO vector $\mathbf{f}_o$, TO vector $\boldsymbol{\tau}_o$

- 1: Initialize: $\bar{K} \leftarrow 0$, $\mathcal{N}_0 = \{\}$
 - 2: **while** $\bar{K} < K$ **do**
 - 3: $\bar{K} \leftarrow \bar{K} + 1$;
 - 4: **Signal Isolation**:
 - 5: Estimate $\hat{\gamma}_{\bar{K}}, \hat{f}_{o,\bar{K}}, \hat{\tau}_{o,\bar{K}}$ using SSE on the residual data $\mathbf{x}_{\mathcal{N}_{\bar{K}-1}}(l)$ corresponding to the known waveform of highest energy;
 - 6: Update $\mathcal{N}_{\bar{K}} \leftarrow \mathcal{N}_{\bar{K}-1} \cup \{(\hat{\gamma}_{\bar{K}}, \hat{f}_{o,\bar{K}}, \hat{\tau}_{o,\bar{K}})\}$;
 - 7: **Local Refinement**:
 - 8: **repeat**
 - 9: Refine all parameters in $\mathcal{N}_{\bar{K}}$. For each $(\gamma_i, f_{o,i}, \tau_{o,i})$ in $\mathcal{N}_{\bar{K}}$, we use $\mathbf{x}_{\mathcal{N}_{\bar{K}} \setminus (\gamma_i, f_{o,i}, \tau_{o,i})}(l)$ as the measurement signal;
 - 10: **for** $i = 1$ to $\bar{K}$ **do**
 - 11: Refine $f_{o,i}$ and $\tau_{o,i}$, and update $\hat{\gamma}_i$;
 - 12: **end for**
 - 13: **until** practical convergence
 - 14: **end while**
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loop, the convergence criterion is the relative change of the log-likelihood functions is below 10^{-6} .

If the number of SOIs is unknown, it can be determined by the Bayesian information criterion (BIC) [26]:

$$\text{BIC}(\tilde{K}) = -2 \ln \mathcal{L}(\hat{\boldsymbol{\chi}}) + 4\tilde{K} \ln(ML), \quad (18)$$

where $\ln \mathcal{L}(\hat{\boldsymbol{\chi}})$ is the log-likelihood function.

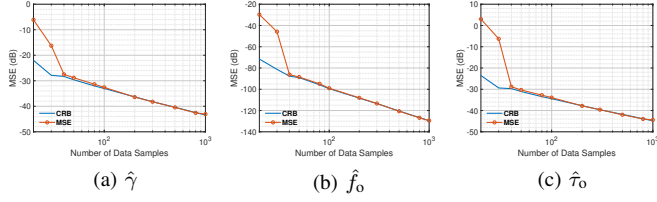


Fig. 1. MSE performance of the proposed SSE and the corresponding CRBs as functions of L for a single SOI.

V. NUMERICAL EXAMPLES

In this section, we evaluate the performance of the proposed estimators for both cases of bi-static and multi-static ISAC systems. Consider a uniform linear array (ULA) with $M = 8$ elements, with inter-element spacing $d = \lambda/2$, where λ denotes the wavelength of the incident signals. The steering vector of the array has the form: $\mathbf{a}(\theta) = [1, e^{-j\frac{2\pi}{\lambda}d \sin(\theta)}, \dots, e^{-j\frac{2\pi}{\lambda}(M-1)d \sin(\theta)}]^T$, where the incident angle θ is relative to the array normal. The incident angles of all SOIs are assumed known. The waveform of each SOI in all cases has the form:

$$s(t) = (1 + e^{j2\pi t/L_1})e^{j2\pi t/L_2}, \quad (19)$$

with L_1 and L_2 being the parameters of the waveforms. We compute the mean squared errors (MSEs) of all estimates based on 500 Monte Carlo trials.

A. Bi-static Case: Spatially colored interference-and-noise

We first consider a single SOI in a bi-static ISAC system. The SOI arrives at the ULA from $\theta = 40^\circ$, with a complex amplitude $\gamma = e^{j\pi/6}$. The normalized CFO is $f_0 = 0.1/\sqrt{2} + 0.1$ and TO is $\tau_0 = 5.2$. The waveform parameters L_1, L_2 are 30, 6, respectively. The interference-and-noise term is a spatially colored circularly symmetric complex-valued Gaussian random vector with zero-mean and an unknown spatial covariance matrix $\mathbf{Q}$. In this example, $\mathbf{Q}$ is assumed to be a Toeplitz matrix with the following form:

$$[\mathbf{Q}]_{m,n} = \rho \cdot (0.99)^{|m-n|} \cdot e^{j\frac{(m-n)\pi}{2}}, \quad (20)$$

where $\rho = 1/\text{SNR}$ and $[\cdot]_{m,n}$ denotes the (m, n) th element of a matrix. Fig. 1 compares the MSEs of complex amplitude, CFO, and TO estimates obtained by the proposed SSE against the Cramer–Rao bounds (CRBs) with SNR set to be 10 dB. The CRBs are derived using the Slepian-Bangs formula [27]. It is observed that the SSE approaches the CRB as the number of snapshots increases. This example demonstrates that the proposed SSE exhibits strong interference suppression capabilities.

B. Multi-static Case: Spatially colored interference-and-noise

We then consider the multiple-SOI case. The parameters of each source are shown in Table I. The interference-and-noise term is the same as that for the bi-static case.

Fig. 2 shows the MSE performance of the proposed ISIR as functions of L with SNR set to be 10 dB. Note that we only

TABLE I
SOURCE PARAMETERS

Source	L_1	L_2	Angle θ	τ_0	f_0	γ
1	30	6	-20°	5.2	$0.01/\sqrt{2}$	$1/2 + \sqrt{3}j/2$
2	23	9	10°	3.5	$-0.01/\sqrt{3}$	$(1+j)/\sqrt{2}$
3	40	11	-40°	4.1	$0.01/\sqrt{5}$	$1/3 + j\frac{2\sqrt{2}}{3}$

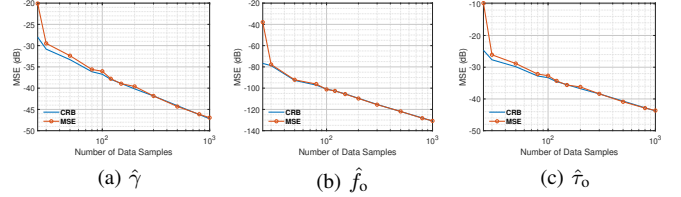


Fig. 2. MSE performance of the proposed ISIR and the corresponding CRBs as functions of L for multiple SOIs when $K = 3$.

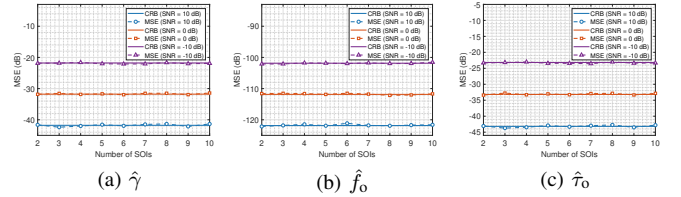


Fig. 3. MSE performance of ISIR and the corresponding CRBs as functions of the number of the SOIs K when $L = 200$.

show the results for the first signal because of the similarities for the other signals. It is evident that the proposed ISIR approaches the CRB as the number of snapshots increases.

Fig. 3 shows the performance of ISIR as functions of the number of SOIs K when SNR=-10,0,10 dB and $L = 200$. The complex amplitudes, CFOs, and TOs are set differently, and their incident angles are separated by 10° . The MSEs of ISIR remain steady as K increases. This implies that ISIR scales well with the number of SOIs.

VI. CONCLUSION

We have studied joint estimation of TO, CFO, and complex amplitude in multi-static ISAC systems in the presence of unknown, spatially colored interference-and-noise. Exploiting known LoS waveforms and array responses, the proposed SSE combines a closed-form complex amplitude update with alternating one-dimensional optimizations for TO and CFO estimation, where FFT-based accelerations make the 1-D searches computationally practical. We further introduce ISIR to handle multiple SOIs, which decouples the high-dimensional search problem into per-SOI subproblems and iterative local refinement while maintaining accuracy. Numerical results show that the proposed methods approach the corresponding CRBs with increasing snapshots, effectively suppress strong interference, and scale well with the number of SOIs, demonstrating reliable synchronization in dense multi-static ISAC deployments.

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